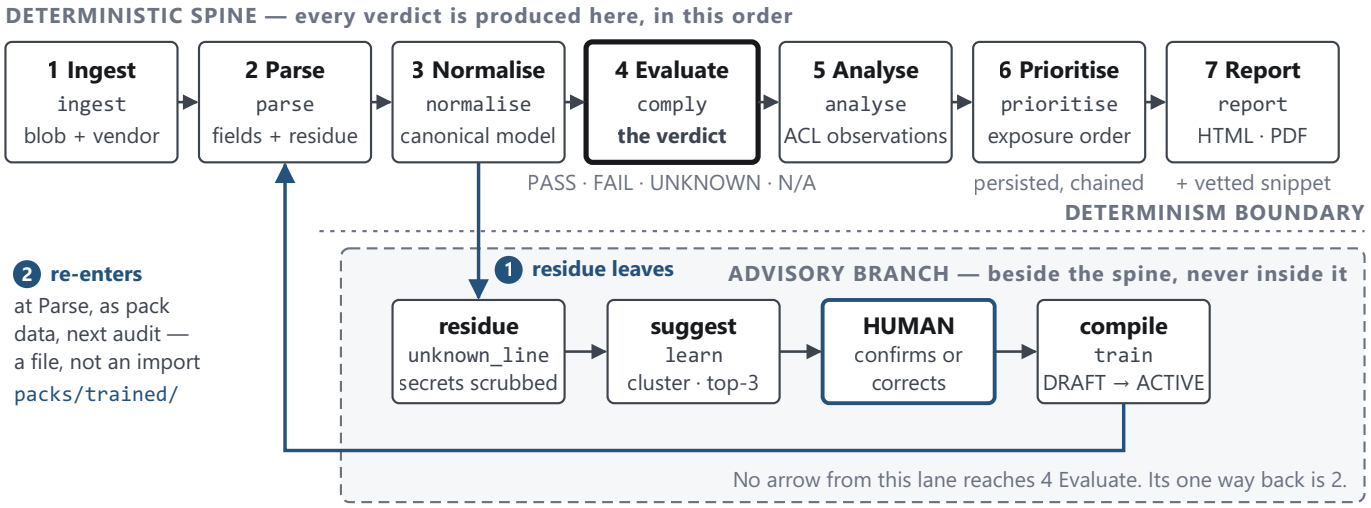
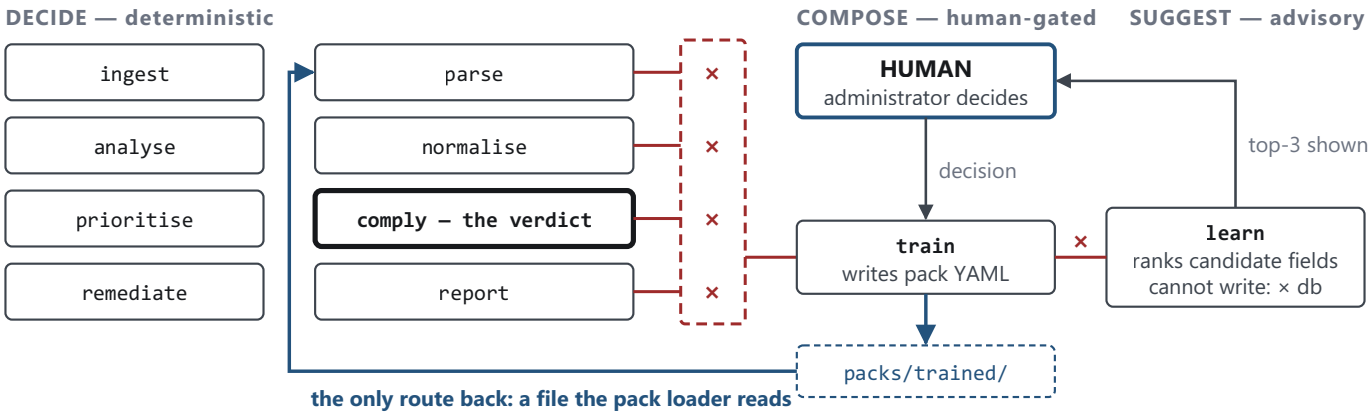
NIRIKSHAK — architecture

AI suggests. Rules decide. Humans confirm.

SIH 2026 · PS 26155 · NTRO · Team Atlantis
Full reference: `docs/architecture.md` · decisions: `docs/adr/`



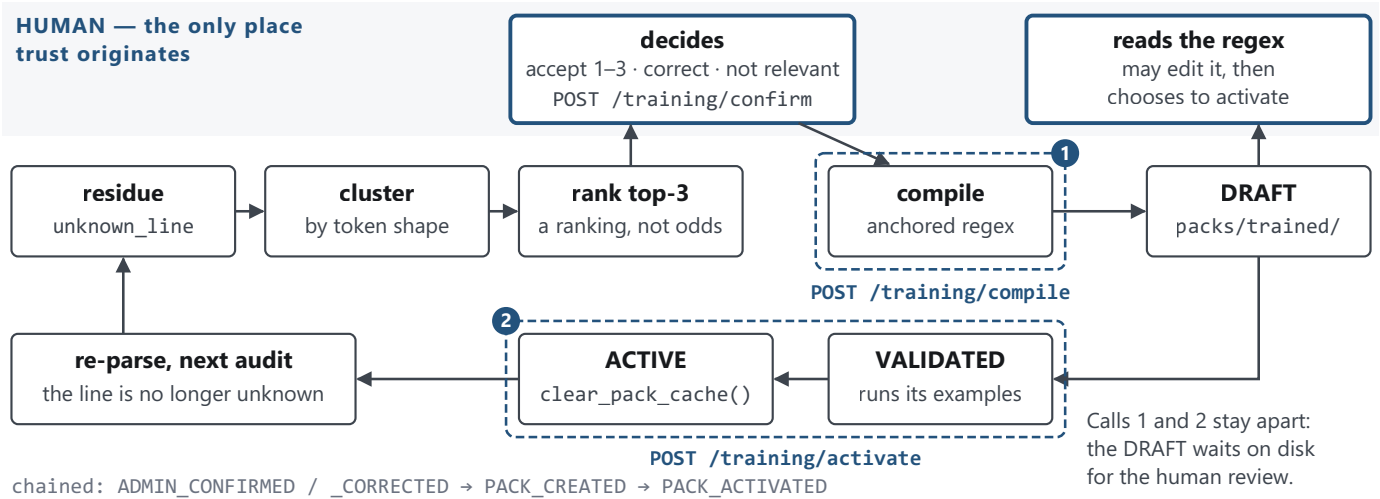
1 · The pipeline. Seven stages in the order `run_audit_endpoint` (`api/routers/audits.py`) calls them. Unmatched lines leave after normalisation (`csm.residue` → `record_residue`); a learned mapping returns only as a vendor-pack file, read by `find_pack`.



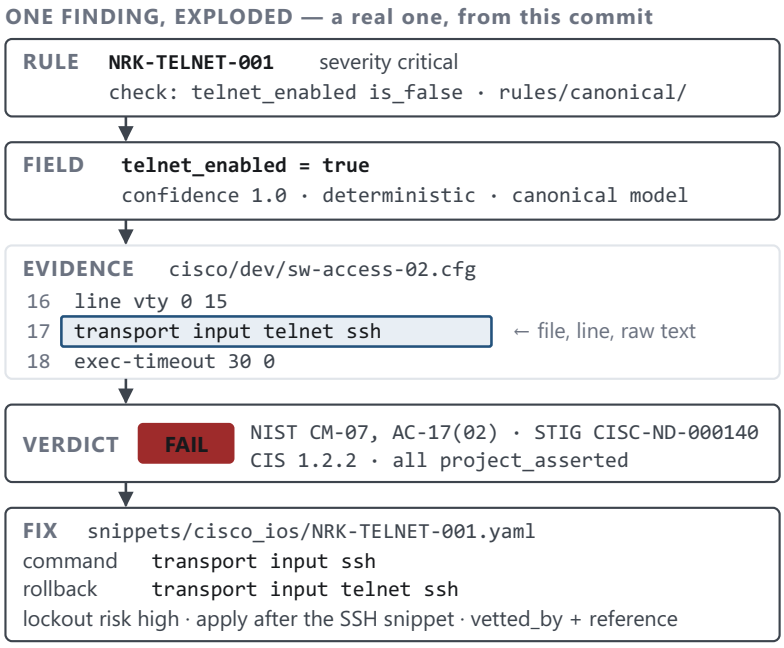
75 forbidden import edges `tests/architecture/test_import_rules.py`
27 of them touch learn or train: 15 point in, 11 point out, 1 joins them (learn × train).

Suggestion has no value `api/models/training.py`	ML imports only in learn `test_learn_boundaries.py`	Model may only suggest `audit/0001_initial.sql`	Uncalibrated → UNKNOWN `api/models/field.py`

2 · The determinism boundary. Each red stub is an import a test forbids: the verdict package cannot see `learn` or `train`, and neither can see it. A suggestion reaches a verdict only as a file that a human caused to exist.



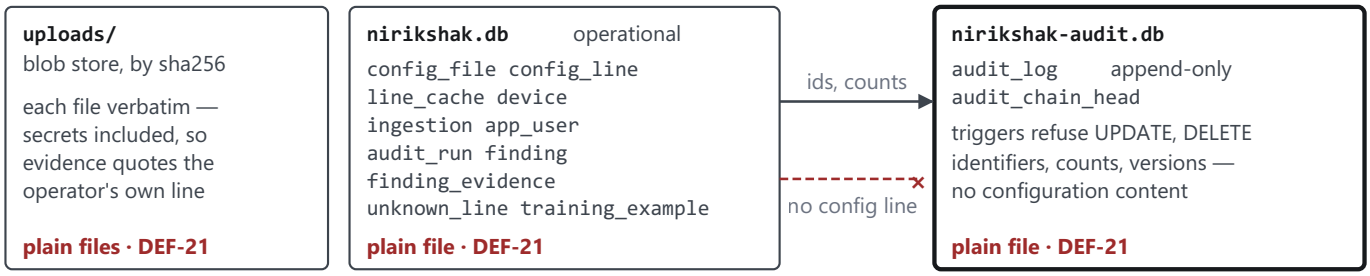
3 · The confirmation loop. `confirm` → `compile_confirmation` → `activate_draft`, all in `api/train/service.py`. The model ranks; an administrator decides, then reads the compiled regex before any configuration is read with it.

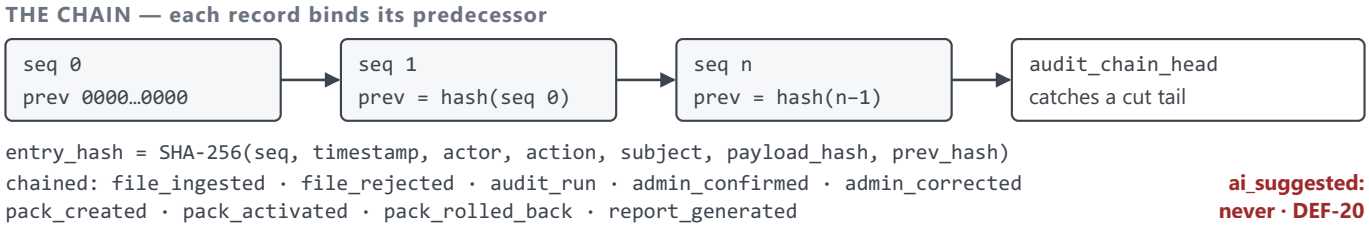




Same run: NRK-SSH-001 is UNKNOWN, cause 2.

4 · Evidence and verdicts. `evaluate_device` (`api/comply/engine.py`) reads only the canonical field; the field carries its line; the fix comes from the vetted library with its rollback and impact. Causes are the eight `UnknownReason` values (`api/models/enums.py`), grouped by where the fix lives; cause 1 is exactly `Finding.needs_training`.





5 · Storage and attestation. Two databases so that "the audit store holds no configuration content" is checked by opening one file (ADR 0009, D4). The chain is tamper-evident, not tamper-proof: it is unkeyed, so an attacker with full write access can recompute it (ADR 0008). Hashing: `compute_entry_hash`, `api/models/audit.py`.

What NIRIKSHAK does not claim — true at commit 28ab52b

- **Encryption at rest.** Required by Rule 6; not built. The databases and `uploads/` are plain files (DEF-21).
- **A chained record of each AI suggestion.** Required by DEVELOPER_GUIDE.md §9; `ai_suggested` is never appended (DEF-20).
- **ISO/IEC 27001.** Not in the selector: a purchased standard, not held. STIG and CIS editions describe Cisco IOS XE 17.x only.
- **An official framework mapping.** Every mapping is `project_asserted`.
- **Calibrated confidence or top-3 accuracy.** No line-level ground truth; calibration refuses under 200 samples.
- **Absence-aware evaluation on real data.** No pack ships a platform default or capability claim.
- **An exposure ranking.** No pack declares management interfaces, so prioritisation abstains.
- **Held-out generalisation.** PAN-OS is sealed; XML and JSON modes raise rather than parse.
- **Field accuracy.** Every corpus file is hand-written by the team; labels are unreviewed.
- **A tamper-proof log.** Tamper-evident only (ADR 0008).
- **Serial numbers.** Not present in a configuration export (ADR 0044).
- **Teaching an unseen vendor from the interface.** The loop extends a platform that has a pack; the first pack is written by hand.
- **FAISS or any LLM.** Ranking is an exact cosine scan (`api/learn/suggest.py`).